

Aleena Varghese

Software Engineer — Full-Stack · APIs · AI/ML

aleenatheresa024@gmail.com · +91 94997 19173
github.com/Aleena24 · linkedin.com/in/aleena-varghese-atv024
Kerala, India · medium.com/@alinavarghese009

SUMMARY

Software engineer who builds **production systems end to end** — full-stack platforms, REST APIs, AI/ML and Google Cloud infrastructure. Founding engineer of **CelerSCET**, a campus-wide ERP (90+ modules, ~500k LOC) running admissions, enrollment, academics and administration in production, with **1,600+ commits and 680+ pull requests merged** as platform lead. MSc in AI/ML; IEEE-published GAN research. Focused on system design, data integrity, automated testing and reliable deployments.

EXPERIENCE

Software Engineer · Full-Stack Developer

Nov 2025 — Present

CelerSCET · Sahrdaya College of Engineering & Technology, Kerala

- **Architected and bootstrapped a campus-wide ERP** serving the full student lifecycle in production — React 18/Vite SPA (PWA), Node.js/Express REST API (~79 route segments), Supabase PostgreSQL with row-level security, deployed on Google Cloud Run.
- **Engineered the admission platform end to end** — multi-step applications, OTP applicant authentication, quota- and degree-aware validation, Razorpay payment integration, QR verification, branded PDF exports and analytics — processing **1,500+ applications and 500+ admissions** in its first cycle.
- **Designed the student provisioning pipeline** — batch creation, class allocation, SR-number generation and institutional email — as a dependency-ordered workflow converting admitted applicants into live student accounts.
- **Implemented platform-wide authorization, session security and shared engines** — a 3-layer permission model with relationally scoped rights (HoD, class teacher, coordinators), enforceable session revocation, registry-driven notifications, a generic approvals engine, and an audit trail with attributable writes and restorable deletes.
- **Own CI/CD and cloud infrastructure**: Cloud Build → Cloud Run (staging + production), Artifact Registry, Cloud Storage, Secret Manager and Cloud Logging; reduced build times from 45+ min to ~4 min and eliminated stale-deploy caching failures.
- **Lead integration for a 5-engineer team** — 680+ PRs reviewed and merged; top contributor to a 275-file automated test suite (~72k lines) that guards system invariants.

AIML Executive · Full-Stack Developer (APIs & AI/ML)

Jan 2025 — Dec 2025

SenseSys Pvt. Ltd. · Bangalore

- **Led an 8-member agile team** as project lead for **Hosting Cloudly**, a cloud hosting platform — developed the React dashboard for uploads, deployment tracking and performance metrics, and integrated REST APIs for authentication, deployment status and log visualization.
- **Directed the ZeroCodeML frontend revamp** — a no-code machine learning platform with automated EDA and model suggestions — building modular React components for model setup, data upload and training progress.
- **Optimized UI load performance by ~30%** through lazy loading, component optimization and asset compression; integrated Supabase-backed auth and data services.

ML Engineer Intern — Prodigy Infotech (Apr–May 2024)

Python Developer Intern — Psyliq (May–Jun 2024)

Graphic Designer — NextU Empower (Sep–Dec 2024)

SELECTED PROJECTS

CelerSCET — Digital Campus ERP (production)

React · Node.js · Supabase/PostgreSQL · GCP

90+ module ERP ecosystem covering admissions → provisioning → academics → administration; detailed under Experience. Live at sahrdaya.celerscet.com.

Hosting Cloudly — Cloud Hosting Platform

React · FastAPI · Supabase · Docker

End-to-end hosting platform (domains, Linux hosting, WordPress deployment) with dashboards for deployment tracking and monitoring; led the 8-member build team.

ZeroCodeML — No-Code Machine Learning Platform

React · Flask · scikit-learn

Upload a CSV and train models without code — automated EDA and model suggestions; led a 3-member team through the V1.0 launch.

HospSurv — Hospital Feedback Platform

React · Supabase

Patient survey system with reward coupons, gamified flows and real-time admin dashboards for response tracking.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Python, SQL, HTML5, CSS3
Frontend	React 18, Vite, React Router, MUI, Tailwind CSS, PWA / service workers
Backend & APIs	Node.js, Express, REST API design, JWT authentication, Flask, FastAPI, Nodemailer, Razorpay
AI / ML	PyTorch, TensorFlow, scikit-learn, GANs, Pandas, NumPy, NLP
Databases	PostgreSQL, Supabase (RLS, Auth, Realtime), Redis, MySQL, SQLite
Cloud & DevOps	Google Cloud (Cloud Run, Cloud Build, Cloud Storage, Artifact Registry, Secret Manager, Cloud Logging), Docker, CI/CD, Git/GitHub, Postman
Practices	Automated testing (node:test), code review, technical documentation, agile delivery

PUBLICATION

Enhancing Satellite Imagery with GAN-based Cloud Removal — IEEE, July 2025

Published in IEEE Xplore · DOI: [10.1109/ICECCC60120.2025.11063844](#) · Presented at ICECCC 2025, CMR Institute of Technology, Bengaluru.

- Engineered a U-Net generator / ResNet discriminator GAN with Perlin-noise cloud synthesis, achieving ~40% improvement in spatial accuracy over baseline models; benchmarked on EuroSAT with PSNR validation and GPU-accelerated training.

EDUCATION

MSc, Artificial Intelligence & Machine Learning	2023 — 2025
CHRIST (Deemed to be University), Bangalore	
Bachelor of Computer Applications	2020 — 2023
Christ College, Rajkot · Outstanding Student Award (CS Dept.)	

LEADERSHIP & LANGUAGES

Project Lead — Hosting Cloudly (8-member) & ZeroCodeML (3-member) teams · Media Head — Gateways 2023 national tech fest (30-member team) · Workshop Facilitator — design & storytelling sessions for 100+ students

Languages: English, Malayalam, Tamil, Hindi, Gujarati

CERTIFICATIONS

Deloitte Data Analytics Job Simulation · AWS Cloud Foundations · Machine Learning with Python · Natural Language Processing · Prompt Engineering for Generative AI · Introduction to Power BI · Data Engineering · Ethical AI